

Joshua Serratelli Schiffman

Vita

Washington, DC, United States of America
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Education

Pennsylvania State University, University Park, PA — *August 2012* Ph.D., Computer Science and Engineering Thesis: *Practical System Integrity Verification in Cloud Computing Environments* Advisor: Trent Jaeger

Pennsylvania State University, University Park, PA — *May 2009* M.S., Computer Science and Engineering Advisor: Trent Jaeger

Pennsylvania State University, University Park, PA — *May 2006* B.S., Computer Science and Engineering with Honors and High Distinction Minors in Mathematics and Japanese Language

Employment

Research Director, Distinguished Technologist — HP Inc., HP Labs, Washington, DC, USA. *May 2023 – Present*

- Research Director and Strategist for HP Labs' Security Lab. Provide technical leadership for HP's supply chain security and lifecycle device management research. In my role as Distinguished Technologist, I offer guidance in navigating strategic engagements with partners, technologies, and global trends in systems security, trusted computing, supply chain security, and secure protocols. I am also focused on building research initiatives between HP and public entities, governments, and researchers.

Master Technologist, Senior Security Researcher — HP Inc., HP Labs, Bristol, England. *May 2015 – Present*

- Senior security researcher at HP Labs Security Lab. Technical leader for endpoint and infrastructure security innovation and research agenda covering supply chain security,

application of trusted computing technology to endpoint device security, user-centric authentication and authorization frameworks, and edge compute infrastructure security. I also lead the HP-wide Security and Privacy Affinity Group to drive excellence and competence in security and privacy at HP.

Member of Technical Staff, Security Architect — Advanced Micro Devices, Inc., Austin, TX. *Sept 2012 – May 2015*

- Design and implementation of hardware and software IP to improve the security and capabilities of AMD products.
- Publish research and submit patents on new security architectures, threats, and mitigations, participate in academic and industry committees, and give talks on AMD security architecture.
- Represent AMD in the Trusted Computing Group (Co-Chair of Mobile Platform Working Group, alternate on Technical Committee), Global Platform, and Cyber Security Research Alliance.
- Aided porting TPM 2.0 reference code to AMD's firmware implementation.

Research Intern — Microsoft Research, Redmond, WA. *Summer 2011*

- Designed and implemented a platform for privacy preserving services. Leveraged Infineon SLE secure hardware, modified Microsoft Hyper-V and designed a Windows Phone 7 application to maintain user privacy in personalized queries. Helped design and evaluate a new multi-client oblivious RAM protocol to protect user privacy in personalized data center services.

Research Intern — Samsung Electronics R&D, San Jose, CA. *Summer 2009*

- Researched distributed cloud computing application security for mobile devices. Designed and implemented an access control manager for sub-delegation of the OAuth web authorization protocol in consumer electronics.

Research Co-op — IBM T. J. Watson Research Center, Hawthorne, NY. *Summer 2008*

- Researched access control policies in virtual machine security and stream computing platforms.

Research Assistant to Trent Jaeger — Pennsylvania State University, University Park, PA. *2006 – 2012*

- Designed, implemented and evaluated a method for installing via CD-ROM network-boot to enable simple verification of the installed filesystem's integrity by leveraging the TPM and late-launch (Intel and AMD) CPU features.
- Designed a secure execution monitor for the Linux kernel in Qtopia on OpenMoko (Neo1973 phone and evaluation board) that prevents untrusted binaries from running under critical SELinux labeled processes.
- Built a runtime integrity monitor for both Xen and Linux KVM that verifies remote client specified integrity policies through a service local to the VM's host. Developed a hardware-based VM introspection mechanism to detect integrity violations in hosted VMs. Evaluated architecture on Eucalyptus and OpenStack cloud platforms.

Technical Intern — Lockheed Martin, King of Prussia, PA. *Summer 2005, 2006*

- Developed web application prototypes for the Coast Guard's Deepwater program. Improved corporate web application for internal requisitions. Automated data entry for the Pennsylvania State Police ArcGIS services.

Publications

For full details, see the [Publications](#) page.

Journals

1. T. Moyer, K. Butler, J. Schiffman, P. McDaniel, and T. Jaeger. Scalable Web Content Attestation. *IEEE Transactions on Computers*, 17, March 2011.
2. Divya Muthukumaran, Joshua Schiffman, Mohamed Hassan, Anuj Sawani, Vikhyath Rao, Trent Jaeger. Protecting the Integrity of Trusted Applications in Mobile Phone Systems. *Security and Communication Networks*, Volume 4, Issue 6, pp 633–650, June 2011.
3. Joshua Schiffman, Trent Jaeger, and Patrick McDaniel. Network-based Root of Trust for Installation. *IEEE Security & Privacy*, Volume 9, Issue 1, pp 40–48, Jan.–Feb. 2011.
4. Trent Jaeger and Joshua Schiffman. Outlook: Cloudy with a Chance of Security Challenges and Improvements. *IEEE Security & Privacy*, Volume 8, Issue 1, pp 77–80, Jan.–Feb. 2010.
5. Lee, K.C.K., Schiffman J., Zheng, B., Lee, W.C., Leong, H.V. Round-Eye: A system for tracking nearest surroundings in moving object environments. *Journal of Systems and Software*, Volume 80, pp 2063–2076, 2007.

Conferences

1. Thalia Laing, Eduard Marin, Mark D. Ryan, Joshua Schiffman, Gaëtan Wattiau. Symbolon: Enabling Flexible Multi-device-based User Authentication. *2022 IEEE Conference on Dependable and Secure Computing (DSC)*, June 2022.
2. Yuqiong Sun, Giuseppe Petracca, Trent Jaeger, Hayawardh Vijayakumar and Joshua Schiffman. CloudArmor: Protecting Cloud Commands from Compromised Cloud Services. *8th IEEE International Conference on Cloud Computing (IEEE CLOUD'15)*, June 2015.
3. Joshua Schiffman, Yuqiong Sun, Hayawardh Vijayakumar, and Trent Jaeger. Cloud Verifier: Verifiable Auditing Service for IaaS Clouds. *2013 IEEE Ninth World Congress on Services (SERVICES '13)*, June 2013.
4. Hayawardh Vijayakumar, Joshua Schiffman, and Trent Jaeger. Process Firewalls: Enforcing Safe Resource Access with Attack-Specific Invariants. *8th ACM European Conference on Computer Systems*, April 2013.
5. J. Lorch, J. Mickens, B. Parno, M. Raykova, J. Schiffman. Toward Practical Private Access to Data Centers via Parallel ORAM. *11th USENIX Conference on File and Storage Technologies*, February 2013.
6. Hayawardh Vijayakumar, Josh Schiffman, and Trent Jaeger. STING: Finding Name Resolution Vulnerabilities in Programs. *21st USENIX Security Symposium (USENIX Security '12)*, August 2012. (19% acceptance rate)
7. Joshua Schiffman, Hayawardh Vijayakumar, Trent Jaeger. Verifying System Integrity by Proxy. *5th International Conference on Trust and Trustworthy Computing*, June 2012.
8. H. Vijayakumar, G. Jakka, S. Rueda, J. Schiffman, and T. Jaeger. Integrity Walls: Finding Attack Surfaces from Mandatory Access Control Policies. *7th ACM Symposium on Information, Computer, and Communications Security (AsiaCCS)*, 2012. (22% acceptance rate)
9. Hayawardh Vijayakumar, Joshua Schiffman, and Trent Jaeger. A Rose by Any Other Name or an Insane Root? Adventures in Namespace Resolution. *7th European Conference on Computer Network Defense*, September 2010. (32% acceptance rate)
10. Patrick Traynor, Joshua Schiffman, Thomas La Porta, Patrick McDaniel, Abhrajit Ghosh, and Farooq Anjum. Constructing Secure Localization Systems with Adjustable Granularity. *IEEE Global Communications Conference (GLOBECOM)*, December 2010. Miami, FL. (35% acceptance rate)

11. Joshua Schiffman, Xinwen Zhang and Simon Gibbs. DAuth: Fine-grained Authorization Delegation for Distributed Web Application Consumers. *POLICY '10: 2010 IEEE International Symposium on Policies for Distributed Systems and Networks*, July 2010. Washington, DC. (19% acceptance rate)
12. Sandra Rueda, Boniface Hicks, Dave King, Thomas Moyer, Joshua Schiffman, Yogesh Sreenivasan, Trent Jaeger, and Patrick McDaniel. An Architecture for Enforcing End-to-End Access Control Over Web Applications. *SACMAT '10: 15th ACM Symposium on Access Control Models and Technologies*, June 2010. Pittsburgh, PA. (24% acceptance rate)
13. Joshua Schiffman, Thomas Moyer, Christopher Shal, Trent Jaeger, and Patrick McDaniel. Justifying Integrity Using a Virtual Machine Verifier. *ACSAC '09: 25th Annual Computer Security Applications Conference*, December 2009. Honolulu, HI. (19% acceptance rate)
14. Thomas Moyer, Kevin Butler, Joshua Schiffman, Patrick McDaniel, and Trent Jaeger. Scalable Web Content Attestation. *ACSAC '09: 25th Annual Computer Security Applications Conference*, December 2009. Honolulu, HI. (19% acceptance rate)
15. Ken C. K. Lee, Josh Schiffman, Baihua Zheng, Wang-chien Lee. Valid Scope Computation for Location-Dependent Spatial Query in Mobile Broadcast Environments. *17th ACM Conference on Information and Knowledge Management*, October 2008. (17% acceptance rate)
16. Divya Muthukumaran, Anuj Sawani, Joshua Schiffman, Brian M. Jung, Trent Jaeger. Measuring Integrity on Mobile Phone Systems. *SACMAT '08: 13th ACM Symposium on Access Control Models and Technologies*, June 2008. (25% acceptance rate)
17. Luke St.Clair, Joshua Schiffman, Trent Jaeger, and Patrick McDaniel. Establishing and Sustaining System Integrity via Root of Trust Installation. *ACSAC '07: 23rd Annual Computer Security Applications Conference*, December 2007. (22% acceptance rate)
18. Ken C. K. Lee, Josh Schiffman, Baihua Zheng, Wang-Chien Lee and Hong Va Leong. Tracking Nearest Surrounders in Moving Object Environments. *IEEE International Conference on Pervasive Services*, 2006.

Workshops

1. Joshua Schiffman and David Kaplan. The SMM Rootkit Revisited: Fun with USB. *2nd International Workshop on Emerging Cyberthreats and Countermeasures (ECTCM '14)*, September 2014.

2. Joshua Schiffman, Thomas Moyer, Hayawardh Vijayakumar, Trent Jaeger, and Patrick McDaniel. Seeding Clouds with Trust Anchors. *2nd ACM Cloud Computing Security Workshop*, October 2010. Chicago, IL.
3. Xinwen Zhang, Joshua Schiffman, Simon Gibbs, Anugeetha Kunjithapatham, and Sangoh Jeong. Securing Elastic Applications on Mobile Devices for Cloud Computing. *1st ACM Cloud Computing Security Workshop*, November 2009. Chicago, IL.
4. William Enck, Sandra Rueda, Yogesh Sreenivasan, Joshua Schiffman, Luke St. Clair, Trent Jaeger, and Patrick McDaniel. Protecting Users from 'Themselves'. *1st ACM Computer Security Architectures Workshop*, November 2007. Alexandria, VA.

Patents

1. Hewlett Packard Development Co LP: Joshua Serratelli Schiffman et al, "Authentication system." U.S. Patent US11144621B2 (October 12, 2021)
2. Hewlett Packard Development Co LP: Joshua Serratelli Schiffman et al, "Managing entitlement." U.S. Patent Application US11146406B2 (October 12, 2021)
3. Hewlett Packard Development Co LP: Joshua Serratelli Schiffman et al, "User equipment to capture identity information from rendering apparatuses." U.S. Patent Application US11137948B2 (October 5, 2021)
4. Hewlett Packard Development Co LP: Joshua Serratelli Schiffman et al, "Establishing shared key data for wireless pairing." U.S. Patent US11057196B2 (July 6, 2021)
5. Hewlett Packard Development Co LP: Joshua Serratelli Schiffman et al, "Distributed authentication." U.S. Patent Application US10771444B2 (September 8, 2020)
6. Hewlett Packard Development Co LP: Joshua Serratelli Schiffman et al, "Content delivery network including mobile devices." U.S. Patent US10558412B2 (February 11, 2020)
7. Advanced Micro Devices Inc.: Joshua Serratelli Schiffman et al, "Protecting state information for virtual machines." U.S. Patent US10152602B2 (December 11, 2018)
8. Advanced Micro Devices Inc.: Joshua Serratelli Schiffman et al, "Method for privileged mode based secure input mechanism." U.S. Patent US9471799B2 (October 18, 2016)
9. Samsung Electronics Co., Ltd.: Joshua Schiffman et al, "Securely Using Service Providers in Elastic Computing Systems and Environments." U.S. Patent Application 20110004916 (April 22, 2010)

Invited Talks

1. Verifying System Integrity by Proxy, *Imperial College London*, September 15, 2014. London, UK.
2. Practical Verification of System Integrity In Cloud Computing Environments, *Trusted Infrastructure Workshop (TIW '13)*, June 5, 2013.
3. Towards Practical Attestation: Challenges and Opportunities, *Trusted Infrastructure Workshop (TIW '12)*, June 10, 2010. Pittsburgh, PA.

Honors

- ACM CCS Conference Student Travel Grant Award 2009
- ACM CCS Workshop Student Travel Grant Awards 2009, 2010
- University Graduate Fellowship Award 2008–2009
- USENIX Association Student Travel Stipend 2007–2011
- IEEE Security and Privacy Travel Grant 2009
- Penn State College of Engineering Fellowship 2006–2007
- ACM SIGMOD Undergraduate Scholarship 2006
- Admitted to the Phi Kappa Phi Honors Society 2006
- Lockheed Martin Engineering Scholars Award 2003

Service

Program Committees

- ACM Symposium on Information, Computer and Communications Security (ASIACCS) — 2014, 2015
- Annual Computer Security Applications Conference (ACSAC) — 2013, 2014
- International Workshop on Emerging Cyberthreats and Countermeasures (ECTCM) — 2015
- ACM Workshop on Scalable Trusted Computing (STC) — 2012
- International Workshop on Security (IWSEC) — 2012

Standards Bodies

- Co-Chair, Trusted Computing Group Technical Committee — June 2020 – Present

Professional Affiliations

- The Association for Computing Machinery (ACM)
- The Institute of Electrical and Electronics Engineers (IEEE)
- USENIX Advanced Computing Systems Association (USENIX)